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ARTICLE

Expanding the scope and scale of the EU's DLT market infrastructures

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Abstract
This article analyses the European Union's regulatory evolution regarding Distributed Ledger Technology (DLT) in financial markets, tracing the journey from the inception of Regulation (EU) 2022/858 to the transformative Market Integration and Supervision Package (MISP). It explores the foundational value proposition of tokenisation, namely atomic settlement and fractionalisation, while contrasting institutional successes like Project Guardian with systemic failures such as the ASX CHESS replacement. The study identifies the 'ceiling on success' inherent in the initial DLT Pilot Regime (DLTR), characterised by restrictive capitalisation thresholds and a lack of native cash leg integration. The analysis further evaluates the 2025 ESMA recommendations and the Commission's subsequent MISP proposal, which seeks to establish a permanent, scalable architecture through unbundled CSD services introducing DLT Notaries and Account Keepers, and significantly elevated aggregate thresholds of €100 billion. The article concludes by arguing that the framework's ultimate success depends on securing European technological sovereignty and maintaining an agile, national-level supervisory model rather than succumbing to premature centralisation.

Keywords DLT pilot regime · Market infrastructure package · Tokenization · Atomic settlement · ESMA · Technological sovereignty · CSDR reform · Crypto-asset service providers · Supervisory convergence

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1 Introduction

The entry into force of Regulation (EU) 2022/858[1] on 23 March 2023 marked a turning point in European capital markets regulation, representing the first formal attempt to reconcile the rigid requirements of traditional market infrastructure with the decentralised nature of Distributed Ledger Technology. Developed as a temporary 'sandbox' for an initial three-year duration, the DLT Pilot Regime (DLTR) was intended to allow market participants and National Competent Authorities (NCAs) to test the trading and settlement of tokenised financial instruments within a controlled environment. However, the initial implementation faced significant structural frictions, leading to a modest adoption rate that necessitated radical legislative change.

This article argues that while the original DLTR was a necessary experimental precursor, its inherent limitations regarding scale, scope, and 'cash leg' integration stifled its commercial viability. Consequently, the proposed amendments within Commission's Market Integration and Supervision Package (MISP)[2] represent a critical evolution toward a more permanent, scalable, and inclusive framework.

The following analysis is structured into three primary sections: Sect. 1 examines the value proposition of tokenisation and refers to a few examples of where DLT projects were a success and failed; Sect. 2 looks into the mechanics and restrictions of the existing 2022 framework, including the assessment and recommendations made by ESMA; the third section details the proposed legislative reforms and their implications for the future of European digital finance. Some concluding remarks are made at the end of the article.

2 Value proposition of tokenisation

The fundamental value proposition of tokenising financial markets lies in the radical optimisation of the post-trade lifecycle, transitioning from sequential reconciliations to a 'golden source' of truth. This shift enables atomic settlement, which effectively collapses the temporal gap between trade execution and finality, thereby reducing counterparty risk and freeing up liquidity that is otherwise trapped in settlement buffers. While the technology is theoretically universal, certain asset classes possess a higher 'tokenisation beta' - sovereign debt and corporate bonds, for instance, are naturally conducive to smart contract automation due to their standardised lifecycles. Conversely, illiquid assets such as private equity and real estate benefit from fractionalisation, which lowers entry barriers and deepens liquidity pools. To realise these benefits, however, the ecosystem requires robust prerequisites: [3] technological interoperability, [2] legal certainty for on-chain property rights, and [1] a functional DLT-native payment facility.[3]

[1] Regulation (EU) 2022/858 of the European Parliament and of the Council of 30.5.2022 on a pilot regime for market infrastructures based on distributed ledger technology, and amending Regulations (EU) No 600/2014 and (EU) No 909/2014 and Directive 2014/65/EU.

[2] European Commission [5].

[3] *Ross, V.* [10].

The tension between innovation and infrastructure is most visible when comparing the successful modularity of Project Guardian and Franklin Templeton against the costly, monolithic failure of the Australian Securities Exchange (ASX) CHESS replacement. Project Guardian, spearheaded by the Monetary Authority of Singapore, demonstrated that DLT could revolutionise wholesale markets by utilising "identity-bound" tokens and automated liquidity pools. Rather than simply digitising a paper trail, the project facilitated atomic settlement and cross-border trading of tokenised bonds and deposits among global giants like JPMorgan and DBS.[4] By utilising Decentralised Finance (DeFi) protocols to automate market making, Project Guardian proved that DLT thrives when it creates new, programmable market structures rather than merely mimicking existing ones. This success is echoed by Franklin Templeton's Benji Investments platform,[5] which scaled a tokenised money market fund to over $1.7 billion in assets by leveraging public blockchains like Polygon and Stellar as the primary record keeping layer. By bypassing the prohibitive costs and latency of traditional transfer agents, Franklin Templeton validated a 'blockchain-native' strategy where the ledger serves as a single source of truth, drastically reducing the reconciliation overhead that historically plagues legacy finance.

In stark contrast, the ASX CHESS replacement project collapsed under the weight of a A$250 million write down, serving as a cautionary tale of technical overreach and architectural misalignment.[6] The project faltered because it attempted to force a distributed ledger, a technology that inherently prioritises consensus and data redundancy, to meet the high frequency, ultra-low latency requirements of a centralised equities market. This friction highlighted that forcing DLT into a monolithic, traditionally governed environment without industry-wide operational readiness creates a 'centralisation paradox'. In this state, the technology is shackled to centralised governance and rigid traditional workflows, causing it to lose its inherent efficiency gains while retaining all of its underlying complexity. Ultimately, these cases suggest that tokenisation succeeds when it is used to disintermediate and automate, but fails when it is treated as a costly, 'bolt-on' upgrade to aging, inflexible infrastructure.

3 The EU's DLT pilot regime

The DLT Pilot Regime[7] establishes a specialised regulatory sandbox by introducing three distinct market infrastructures: [a] the DLT Multilateral Trading Facility (MTF), [b] the DLT Settlement System (SS), and the hybrid [c] DLT Trading and Settlement System (TSS). The TSS model is particularly transformative, as it permits the vertical integration of execution and post trade processing within a single operational framework. To incentivise the adoption of these models, the regime provides targeted exemptions from established MiFID II/MiFIR and CSDR mandates. These

[4] Monetary Authority of Singapore [9].

[5] Franklin Templeton [8].

[6] DigFin [4].

[7] European Securities and Markets Authority [7].

derogations providing among others for direct retail participation by bypassing traditional intermediaries, allow for the replacement of conventional book entry accounts with distributed ledgers, and relax some of the settlement discipline requirements in recognition of the technology's capacity for near instant finality.

The operational core of the DLTR is defined by its capacity to offer 'exemptive relief' from segments of the Union's financial services framework that are conceptually incompatible with decentralised architectures. By acknowledging that existing regulations were not designed for distributed ledgers, the regime establishes a tailored legal environment for 'tokenised securities', specifically digital representations of MiFID II financial instruments. Crucially, this scope is distinct from the broader crypto asset ecosystem, the DLTR does not extend to asset referenced tokens (ARTs) or electronic money tokens (EMTs), which remain governed by the Markets in Crypto Assets Regulation (MiCA). This clear demarcation ensures that the pilot remains focused on traditional securities while testing the efficacy of the underlying technology.

Operationalising these infrastructures requires 'specific permission' from a National Competent Authority (NCA), which functions as a six-year supplement to a firm's standard investment firm or Central Securities Depository (CSD) authorisation. This dual-track application process allows market entrants to reconcile traditional regulatory safeguards with the flexibility of DLT, provided they satisfy rigorous organisational requirements and risk mitigation protocols. To prevent systemic contagion during this experimental phase, the regime is primarily restricted to small cap shares and specific debt instruments. Furthermore, participants are mandated to maintain a robust transition strategy, ensuring a safe migration of all financial instruments back to legacy systems should the platform exceed its aggregate market value thresholds or conclude its authorised duration. This mechanism ensures that while the permission provides a stable multi year window for innovation, the integrity of the wider financial system remains protected through mandatory exit readiness.

Despite these gradual advancements, the regime's utility is significantly constrained by rigid caps, specifically €500 million for equities and €1 billion for bonds, which create a 'ceiling on success' that frequently deters institutional involvement. The requirement to trigger a transition strategy once the total market capitalisation reaches €9 billion effectively penalises scale, forcing growing platforms to revert to traditional infrastructures that are often technologically incompatible with native DLT operations. This systemic friction is further exacerbated by the lack of a native DLT cash leg. By remaining dependent on off chain commercial bank money, the efficiency of Delivery versus Payment (DvP) is compromised, leaving these infrastructures functionally isolated from the broader, real-time financial ecosystem.

Under the DLT Pilot Regime, National Competent Authorities (NCAs) are the primary supervisors since NCAs grant, monitor and withdraw specific permissions to the applicant entities, while ESMA has an important coordination and convergence role. In particular the role of ESMA consists of (i) issuing the non-binding opinions before an NCA grants permission; (ii) drafting reports on functioning of the regime; (iii) coordinating NCA practices; (iv) drafting Guidelines and Q&A. In addition, ESMA publishes on its website the list of DLT market infrastructures.[8] Currently only 6 en-

[8]Under its mandates in Articles 8(11), 9(11) and 10(11) of the DLT Pilot Regime, ESMA is required to publish the list of all authorised DLT Market Infrastructures, the start and end dates of their spe-

tities operate under the DLT Pilot Regime, 4 as DLT TSS, 1 as DLT MTF and 1 as DLT CSD.

In June 2025 ESMA issued a report on the functioning and review of the DLT Pilot Regime[9] which *inter alia* criticised the framework and made a number of recommendations. ESMA's assessment identified the imperative to transition the DLT Pilot Regime from a temporary framework into a permanent regulatory fixture by eliminating the current six-year 'sunset clause'. This shift would provide the legal certainty necessary to catalyse long-term institutional investment and mitigate the 'early adopter risk' that currently stifles market entry. To address the 'ceiling on success' created by existing volume constraints, ESMA advocated for the introduction of tiered and adjustable thresholds. Rather than adhering to rigid, *ex-ante* legislative limits, NCAs would be empowered to apply a principle of proportionality, tailoring thresholds to an entity's specific risk profile and business model. This dynamic calibration, potentially overseen by ESMA to ensure supervisory convergence across Member States, would seek to balance the need for meaningful commercial scale with the preservation of market integrity.

A significant recommendation made by ESMA involves broadening the scope of eligible financial instruments to include complex and illiquid assets, such as structured products and Alternative Investment Funds (AIFs). ESMA noted that the inherent automation of smart contracts is particularly well suited for these instruments, however, such expansion necessitates a corresponding reinforcement of investor protection frameworks. To prevent retail exposure to undue risk, the report suggested a segmented access model where complex DLT native products are restricted to professional or sophisticated investors. Furthermore, while acknowledging the current necessity of settlement via commercial bank money or Electronic Money Tokens (EMTs), ESMA reiterated that central bank money remains the 'preferred settlement asset'. Consequently, it recommended that infrastructures utilising private money settlement remain subject to more stringent risk-based limits until DLT compatible central bank solutions become operational.

In the long term, ESMA criticised the current requirement for DLT Market Infrastructures (MIs) to 'wind down' operations upon exceeding growth thresholds, labelling it counterproductive for the ecosystem. The report proposed a 'graduation pathway' wherein successful entities can transition into a permanent regulatory regime, potentially by embedding DLT specific provisions directly into sectoral frameworks like MiFID II or CSDR. A critical component of this evolution is the formalisation of the DLT Trading and Settlement System (TSS). ESMA suggested that the unique capacity for 'atomic settlement' within the TSS model may require a shift toward functional, outcome-based regulation, focusing on risk controls and settlement finality rather than traditional, siloed institutional roles. This approach would aim to reconcile the 'same risk, same rules' principle with the distinctive operational realities of decentralised technology, ultimately integrating DLT into the mainstream financial architecture of the European Union.

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cific permissions, the list of exemptions granted to each of them, and any lower thresholds set by the competent authorities. https://www.esma.europa.eu/sites/default/files/2026-01/Authorised_DLT_Market_Infrastructures.pdf, accessed 8 April 2026.

[9] European Securities and Markets Authority [6].

4 The EU commission' proposal – markets infrastructure supervision package

In response to the barriers identified during the initial pilot phase, the European Commission's proposed amendments under the MISP signify a fundamental redesign of the DLT architecture.[10] This shift represents a decisive move away from a transient 'sandbox' mentality toward a more robust, flexible, and permanent market structure. The most striking evolution is the exponential increase in operational scale. Standard aggregate market value thresholds are set to rise to €100 billion, while the previously restrictive product specific thresholds are abolished to allow market forces to dictate asset composition more freely. To foster a tiered ecosystem, a new Simplified Regime is introduced under Article 7a for smaller infrastructures, such as those operated by CASPs, investment firms, or specialised CSDs, provided they remain under a €10 billion total market value cap. This tiered approach is supported by foundational changes to sectoral legislation, most notably via amendments to Article 2 of the CSDR, which modernise the legal definitions of 'book-entry', 'securities accounts', and 'cash' to natively accommodate DLT and e-money tokens (EMTs).

Architecturally, the proposal introduces a modular unbundling of traditional functions, moving beyond the monolithic CSD model. New provisions under Articles 10a to 10f allow for service specific regulation of DLT Notary and Central Maintenance services. Critically, these core services can now be provided by a broader range of entities, including among others regulated markets and CASPs, enabling the issuance and recording of DLT financial instruments outside the traditional CSD perimeter, provided they are settled within a regulated infrastructure. This is complemented by a new settlement model involving DLT Account Keepers who operate within previously authorised Settlement Schemes. These schemes, which must be assessed by ESMA, facilitate the settlement of transactions between account keepers using central bank money or authorised EMTs. By amending Title IV of the CSDR to permit settlement via e-money tokens authorised under MiCA, the reform finally addresses partially the 'cash leg' deficit, enabling true on chain settlement finality and resolving the Delivery versus Payment (DvP) frictions that hampered earlier iterations.

The scope of the regime has been further expanded to include DLT Trading Venues (DLT TV), a designation that reflects the newfound ability to operate both multilateral trading systems and organised trading facilities (OTFs) within the pilot. Under Article 4, the operator definition is broadened to include CASPs authorised to operate trading platforms, effectively bridging the gap between crypto native innovation and the traditional financial system. To manage the heightened complexity of these vertically integrated or unbundled structures, new Articles 4a and 5a introduce the possibility for operators to obtain specific exemptions from MiFID II, MiFIR, and CSDR provisions where existing rules are found to be technologically incompatible. However, this flexibility is balanced by enhanced supervisory rigour, the proposal mandates the establishment of supervisory colleges for those infrastructures that present a cross European footprint and requires NCAs to seek non-binding opinions from ESMA before granting derogations. Furthermore, Article 45a introduces targeted requirements

[10] European Commission [5].

for managing risks inherent to DLT when utilised outside of standard outsourcing arrangements, ensuring that the migration to decentralised ledgers does not bypass fundamental safety standards.

The regime modifies the role of NCAs and ESMA, NCAs retain supervisory role for individual DLT market infrastructures; however, ESMA's role is substantially enhanced. The proposal significantly expand the role of ESMA, notably with (i) more active involvement in assessment and approval processes; (ii) power to adjust regime thresholds to promote convergence; (iii) enhanced role in monitoring systemic risks; (iv) development of regulatory and implementing technical standards; and (v) full direct supervision of CASPs under MiCA transferred from NCAs to ESMA.[11]

Ultimately, the transition from the original DLTR to the MISP reflects a strategic realisation that for DLT to achieve systemic relevance, it must move beyond isolated experiments and into the core of market liquidity. The new framework successfully addresses the primary 'drags' on adoption, namely scale, scope, and payment diversity, while introducing lighter, more proportionate requirements for specialised settlement activities. To ensure that this expanded ecosystem does not become fragmented, Article 10g mandates that technical standards are adopted to support interoperability between diverse DLT infrastructures, with ESMA providing ongoing technical advice to the Commission on these standards. While DLT's advantages in transparency and cost reduction are increasingly tangible, the complexity of this updated drafting suggests that the path to a fully integrated digital capital market will require continuous refinement. The success of this evolution will depend on the industry's ability to leverage these new modular roles without inadvertently creating unmanaged systemic vulnerabilities in the removal of traditional intermediaries.

5 Conclusion

The transition toward a tokenised financial ecosystem offers the European Union a historic opportunity to redefine the architecture of its capital markets and advance the objectives of the Savings and Investments Union. By leveraging Distributed Ledger Technology (DLT) the Union can modernise its 'financial plumbing' and enhance its global competitiveness. However, the successful evolution of this framework must be underpinned by a commitment to European technological sovereignty. In the absence of concrete actions to secure indigenous infrastructure and localised data storage, the Union's digital financial architecture remains structurally vulnerable to the extra territorial reach and operational dependencies of third-country providers. Ensuring that the technological 'rails' of the tokenised economy are anchored within the EU is not merely a technical requirement but a geopolitical necessity for achieving strategic autonomy.[12]

The success of this framework ultimately depends not just on how well the legislation is written, but on how innovation is supervised in practice. The MIP's proposals

[11] The following two articles, by one of the co-authors, counterargue the centralisation of supervision of CASPs as proposed in the MISP: *Buttigieg, Gauci* [2]; *Buttigieg, Gauci* [3].

[12] *Buttigieg* [1].

to centralise direct supervision of CASPs, major CSDs, and large trading venues under ESMA mark a significant shift, one that deserves careful scrutiny. Centralised oversight is a legitimate tool against regulatory fragmentation, but it carries the risk of distancing supervisors from the local technological ecosystems where these projects actually take root and evolve.

The better model is one where supervisory proximity keeps pace with the development of DLT financial market infrastructures. The current architecture (national supervision guided by ESMA coordination) broadly works, but it needs to go further. ESMA should play a stronger coordination and convergence role in shaping truly common supervisory expectations, both for the initial granting of special permissions under the DLT Pilot Regime, and for the ongoing oversight of DLT FMIs. A common supervisory handbook, built at ESMA level and implemented consistently by NCAs on the ground, would strike the right balance between harmonisation and subsidiarity.

The EU's priority should be to strike the right balance, building harmonised, coherent practices across Member States while preserving the supervisory proximity that serves both market stability and investor protection. Consolidating power at the centre before the technology has reached institutional maturity would be premature and potentially counterproductive. Paired with a firm commitment to technological sovereignty, this approach would keep EU capital markets both attractive and competitive on the global stage.

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Declarations

Competing interests The authors are high level officials of financial regulatory authorities in their jurisdiction and contribute actively in discussions at European level.

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